

Statistics - Set 1

TMUA practice questions by topic. Specification M6.1-M6.4. 11 questions.

Each question has one correct answer. No calculator. Answers and worked solutions are in the separate solutions PDF.

- 1 The mean of five numbers is 12. When a sixth number is added the mean becomes 14. What is the sixth number?
- A** 24
B 84
C 20
D 26
E 16
F 14
- 2 A histogram has a bar for the class $10 \leq x < 15$ of height (frequency density) 3.2 and a bar for $15 \leq x < 25$ of height 1.5. What is the total frequency of these two classes?
- A** 15
B 23.5
C 31
D 4.7
E 16
F 47
- 3 The numbers 3, 7, x , 12, 18 (with $7 \leq x \leq 12$) have mean equal to median. Find x .
- A** 12
B 7
C 10
D 8
E 11
F 9

- 4 The table shows grouped data: $0 \leq t < 10$: frequency 4; $10 \leq t < 20$: frequency 10; $20 \leq t < 40$: frequency 6. Estimate the mean.
- A 16
 - B 18
 - C 20
 - D 15
 - E 22
 - F 17.5
- 5 Which of the following statements about a scatter graph showing strong positive correlation between hours of revision and exam mark is/are valid?
- I Students who revised more tended to score higher.
 - II Revising more causes a higher mark.
 - III A line of best fit could be used to estimate the mark of a student who revised for a number of hours within the range of the data.
- A none of them
 - B I only
 - C II only
 - D III only
 - E I and II only
 - F I and III only
 - G II and III only
 - H I, II and III
- 6 A set of n numbers has mean m . Each number is doubled and then 3 is added. Which of the following is the new mean, and which the new range (if the original range was R)?
- A mean $2m + 3$, range $2R$
 - B mean $2m$, range $2R$
 - C mean $m + 3$, range R
 - D mean $2m + 3$, range $2R + 3$
 - E mean $2m + 3$, range R

- 7 A cumulative frequency graph for 80 observations passes through (20, 20), (30, 40) and (45, 60). Estimate the interquartile range.
- A 40
 - B 10
 - C 20
 - D 25
 - E 15
 - F 30
- 8 Class A has 20 students with mean mark 60; class B has 30 students with mean mark 70. What is the mean mark of all 50 students?
- A 68
 - B 65
 - C 62
 - D 67
 - E 64
 - F 66
- 9 For which of the following data sets is the mean greater than the median?
- A 1, 2, 3, 4, 5
 - B 5, 5, 5, 5, 5
 - C 2, 4, 6, 8, 10
 - D 1, 2, 3, 4, 20
 - E 1, 10, 11, 12, 13
- 10 The median of 7 distinct integers is 10 and the mean is 12. The smallest number is increased by 20. Which of the following must be true?
- A The mean and median both increase.
 - B The median increases and the mean does not change.
 - C The mean increases and the median does not change.
 - D The mean increases and the median may or may not change.
 - E Neither the mean nor the median changes.

- 11** A set of five distinct positive integers has median 10 and mean 10. Which of the following statements must be true?
- I** The largest value is at least 12.
 - II** The smallest value is at most 8.
 - III** The range is at least 4.
- A** none of them
- B** I only
- C** II only
- D** III only
- E** I and II only
- F** I and III only
- G** II and III only
- H** I, II and III